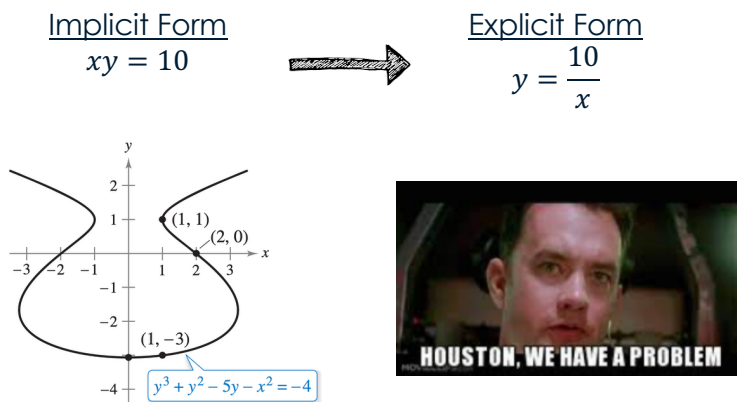




3.2 Implicit Differentiation

Implicitly Defined Relations

In mathematics, an implicitly defined relation is a relationship between variables that is not explicitly solved for one variable in terms of the other(s). Sometimes an implicitly defined relation can be rewritten explicitly.



To differentiate an implicitly defined relation that cannot be rewritten explicitly use implicit differentiation.

Chain Rule and Implicit Differentiation

In Terms of x	In Terms of y
$\frac{d}{dx}(x^2) = 2x$	$\frac{d}{dx}(y^2) = 2x \frac{dy}{dx}$
$\frac{d}{dx}(\sin x) = \cos x$	$\frac{d}{dx}(\sin y) = \cos y \frac{dy}{dx}$
$\frac{d}{dx}(\ln x) = \frac{1}{x}$	$\frac{d}{dx}(\ln y) = \frac{1}{y} \frac{dy}{dx}$
$\frac{d}{dx}(e^{3x}) = 3e^{3x}$	$\frac{d}{dx}(e^{3y}) = 3e^{3y} \frac{dy}{dx}$

Ex 1: Find $\frac{dy}{dx}$ implicitly.

a) $x^2 - y^2 = 16$

$$\frac{dy}{dx} = \frac{x}{y}$$

b) $y^3 + y^2 - 5y - x^2 = -4$

$$\frac{dy}{dx} = \frac{2x}{3y^2 + 2y - 5}$$



c) $x^2y - 2 \cos 3x = 3$

$$\frac{dy}{dx} = \frac{-2xy - 6 \sin 3x}{x^2}$$

d) $\sin^3 x + \cos^4 y = 4$

$$\frac{dy}{dx} = \frac{3 \sin^2 x \cos x}{4 \cos^3 y \sin y}$$

Ex 2:

If $\ln(2x + y) = x + 1$, then $\frac{dy}{dx} =$

(A) -2

(B) $2x + y - 2$

(C) $2x + y$

(D) $4x + 2y - 2$

(E) $y - \frac{y}{x}$

Ex 3: Consider the relation $x^2 - xy + y^2 = 7$

a) Find $\frac{dy}{dx}$ implicitly.

$$\frac{dy}{dx} = \frac{y - 2x}{2y - x}$$

b) Find the slope at the point $(-1, 2)$.

$$\left. \frac{dy}{dx} \right|_{(-1,2)} = \frac{4}{5}$$

c) Write an equation of the tangent line to the graph at the point $(-1, 2)$.

$$y - 2 = \frac{4}{5}(x + 1)$$

d) Write an equation of the normal line to the graph at the point $(-1, 2)$.

$$y - 2 = -\frac{5}{4}(x + 1)$$



Ex 4: Consider the relation $4x^2 + y^2 - 8x + 4y + 4 = 0$. At what points, if any, does the relation have a horizontal tangent line?

The relation has a horizontal tangent when

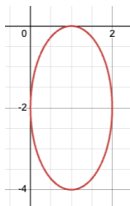
$$\frac{dy}{dx} = 0.$$

$$\frac{dy}{dx} = \frac{4 - 4x}{y + 2} = 0 \Rightarrow x = 1$$

(1, 0) and (1, -4)



***Teaching Tip:** Students often times stop after finding $x = 1$. Remind the students when finding points, they will need to utilize the original equation.



This problem lends nicely to connecting representations. Show students the graph of the relation using Desmos. Students can then make sense of their answers.

Ex 5: **2025 AB6 – parts A. and C.**

Consider the curve G defined by the equation $y^3 - y^2 - y + \frac{1}{4}x^2 = 0$.

A. Show that $\frac{dy}{dx} = \frac{-x}{2(3y^2 - 2y - 1)}$.

SCORING STATS!
Guess the mean: **3.6/9**

C. For $x > 0$ and $y > 0$, there is a point S on the curve G at which the line tangent to the curve at that point is vertical. Find the y -coordinate of point S . Show the work that leads to your answer.

A.

$$\frac{d}{dx} \left(y^3 - y^2 - y + \frac{1}{4}x^2 \right) = \frac{d}{dx}(0)$$

$$\Rightarrow 3y^2 \cdot \frac{dy}{dx} - 2y \cdot \frac{dy}{dx} - \frac{dy}{dx} + \frac{x}{2} = 0$$

Implicit differentiation

Point 1 (P1)

$$\Rightarrow (3y^2 - 2y - 1) \frac{dy}{dx} = -\frac{x}{2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{-x}{2(3y^2 - 2y - 1)}$$

Verification

Point 2 (P2)

C.

For $x > 0$, the curve G has a vertical tangent line when $2(3y^2 - 2y - 1) = 0$.

Sets denominator equal to 0

Point 5 (P5)

$$2(3y^2 - 2y - 1) = 0 \Rightarrow 2(3y + 1)(y - 1) = 0$$

Answer

Point 6 (P6)

Because $y > 0$, it follows that $y = 1$.



Higher Order Derivatives

Higher order derivatives refer to the repeated process of differentiating a function multiple times. These derivatives have applications in various fields like physics (analyzing motion) and economics (cost analysis).

	Lagrange	Leibniz
Function	$f(x)$	y
1 st Derivative	$f'(x)$	$\frac{dy}{dx}$
2 nd Derivative	$f''(x)$	$\frac{d^2y}{dx^2}$
3 rd Derivative	$f'''(x)$	$\frac{d^3y}{dx^3}$
4 th Derivative	$f^{(4)}(x)$	$\frac{d^4y}{dx^4}$
n th Derivative	$f^{(n)}(x)$	$\frac{d^ny}{dx^n}$



Careful! $f^{(4)}(x) \neq f^4(x)$.
 $f^{(4)}(x)$ is the 4th derivative of f
while $f^4(x)$ is the 4th power of f .

Ex 6: If $x^2 + y^2 = 1$ Find $\frac{d^2y}{dx^2}$ implicitly.

$$\frac{d^2y}{dx^2} = \frac{-1}{y^3}$$

Ex 7:

Let $y = f(x)$ be a twice-differentiable function such that $f(1) = 2$ and $\frac{dy}{dx} = y^3 + 3$. What is the value of $\frac{d^2y}{dx^2}$ at $x = 1$?

- (A) 12 (B) 66 (C) 132 (D) 165

Ex 8:

If $(x + 2y) \cdot \frac{dy}{dx} = 2x - y$, what is the value of $\frac{d^2y}{dx^2}$ at the point $(3, 0)$?

- (A) $-\frac{10}{3}$ (B) 0 (C) 2 (D) $\frac{10}{3}$ (E) Undefined